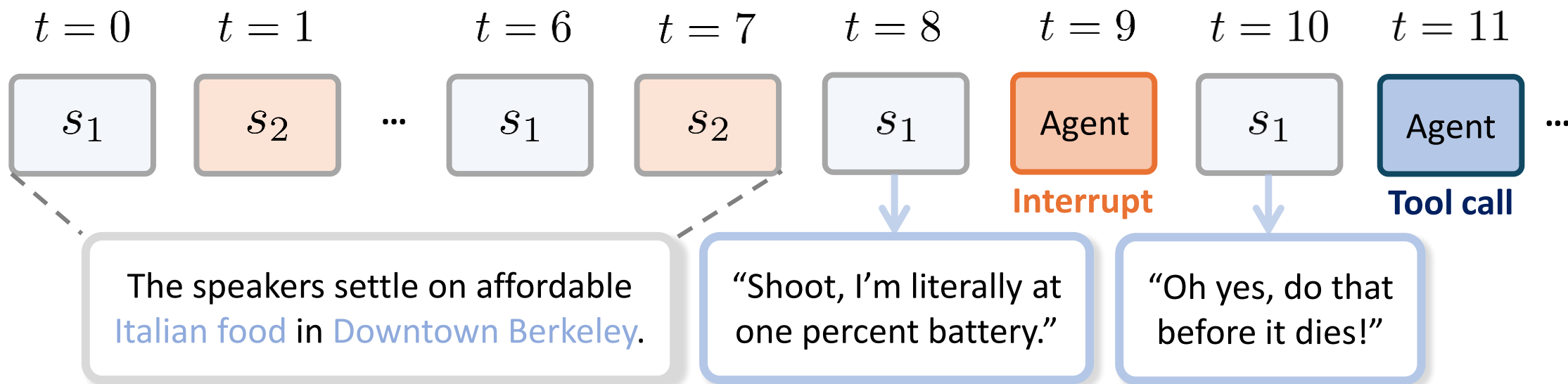


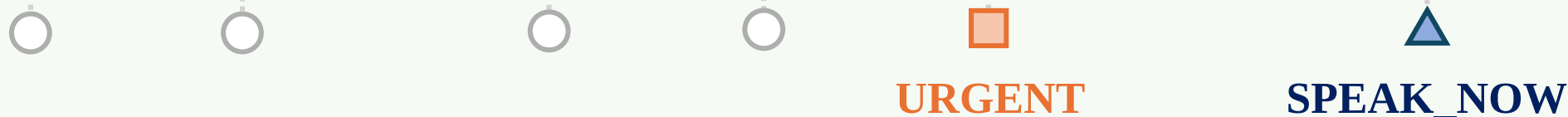
Dialogue Setting

$$u_t = (s_t, x_t, a_t)$$



Stage 1 (When to act)

$$\pi_{\theta} : W_{1:t} \rightarrow (\hat{b}_t, \hat{r}_t)$$



Intervention Labels :

○ SILENT ▲ SPEAK_NOW ■ URGENT ◆ SPEAK_BRIEF_AT_END

Stage 2 (What to call)

$$\phi_{\theta} : (W_{1:t}, \mathcal{C}) \rightarrow (\tau, \mathbf{p})$$

Input $\mathcal{C}$: Candidate Tools

Open Data Minneapolis.getStatistics
University of Oslo.getTeacherInfo
▶ Yelp.searchBusiness

Output $(\tau, \mathbf{p})$: Tool Invocation

τ Yelp.searchBusiness
 $\mathbf{p}$ { term: "**Italian**",
location: "**Downtown Berkeley, CA**", ... }